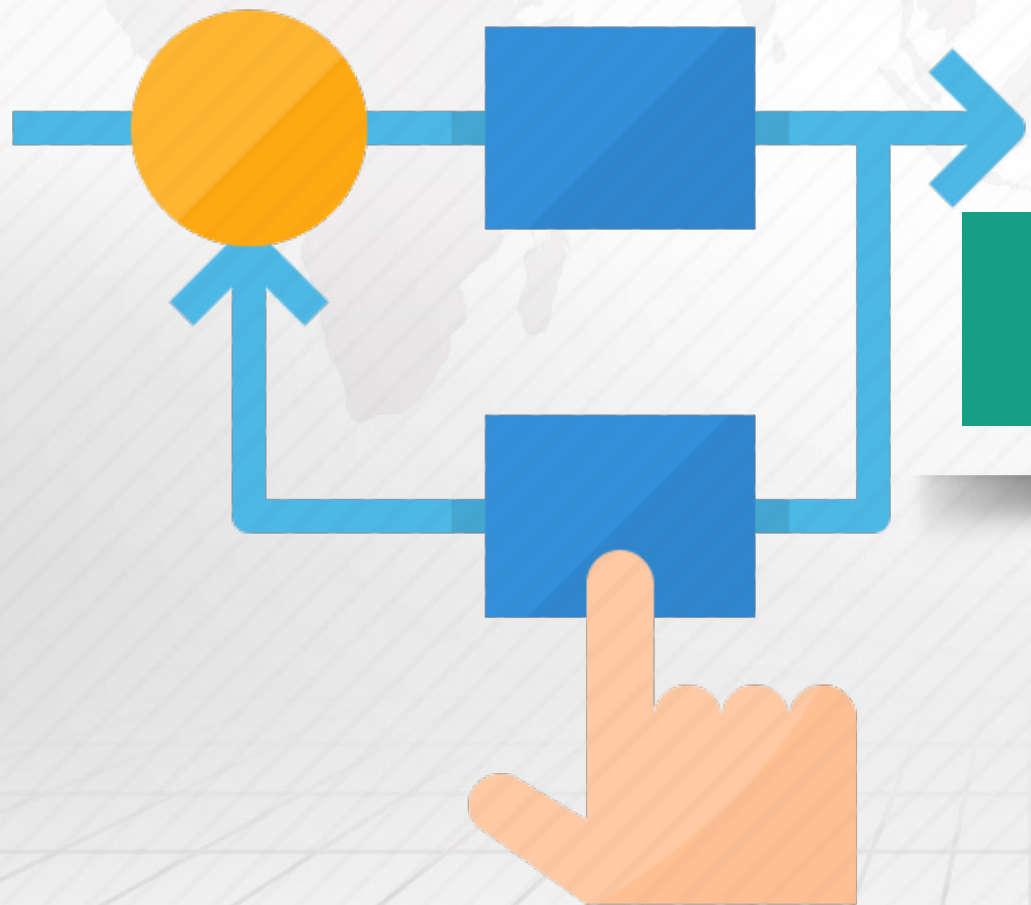




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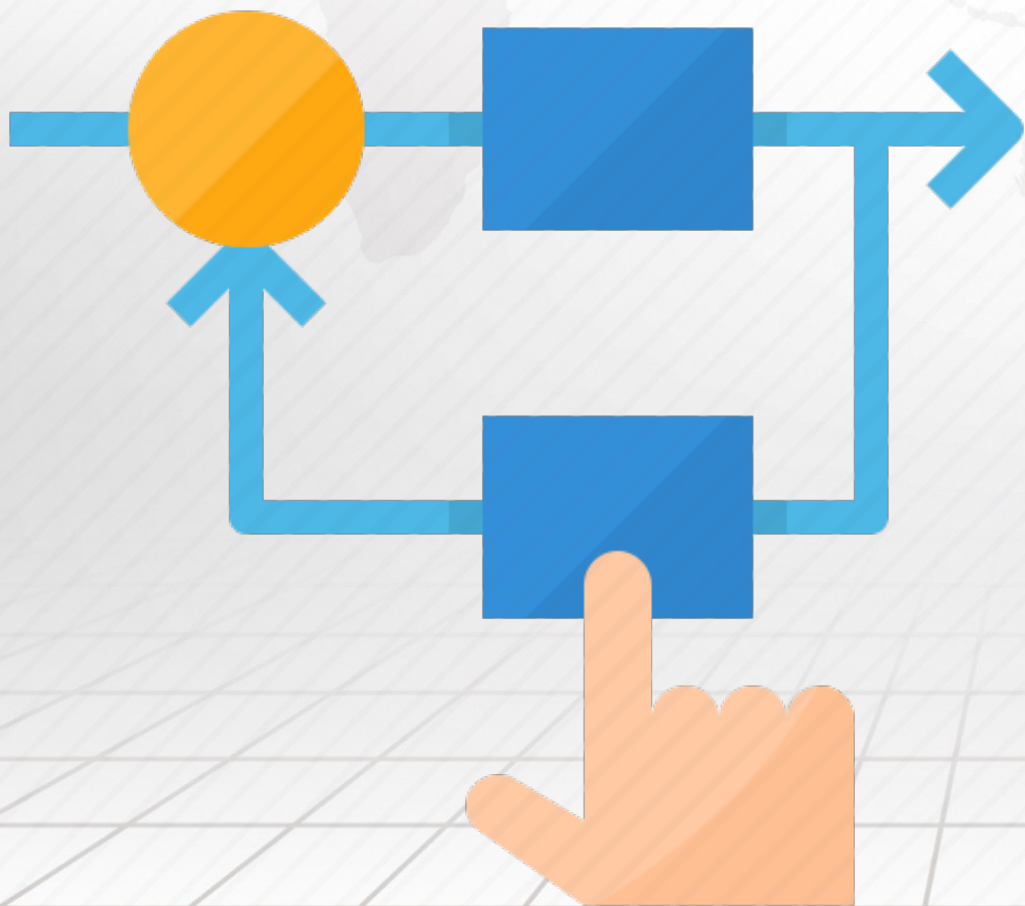


自动控制原理

第一章	第二章	第三章	第四章	第五章	第六章
基本概念	数学模型	时域分析法	根轨迹法	频域分析法	校正方法

23. 滞后超前：双管齐下

学习目标



01

能解释滞后超前网络典型特性

重

能运用滞后超前校正设计控制器

难

能综合运用计算机辅助分析系统

04

能了解滞后超前的运用场景

滞后-超前网络特性

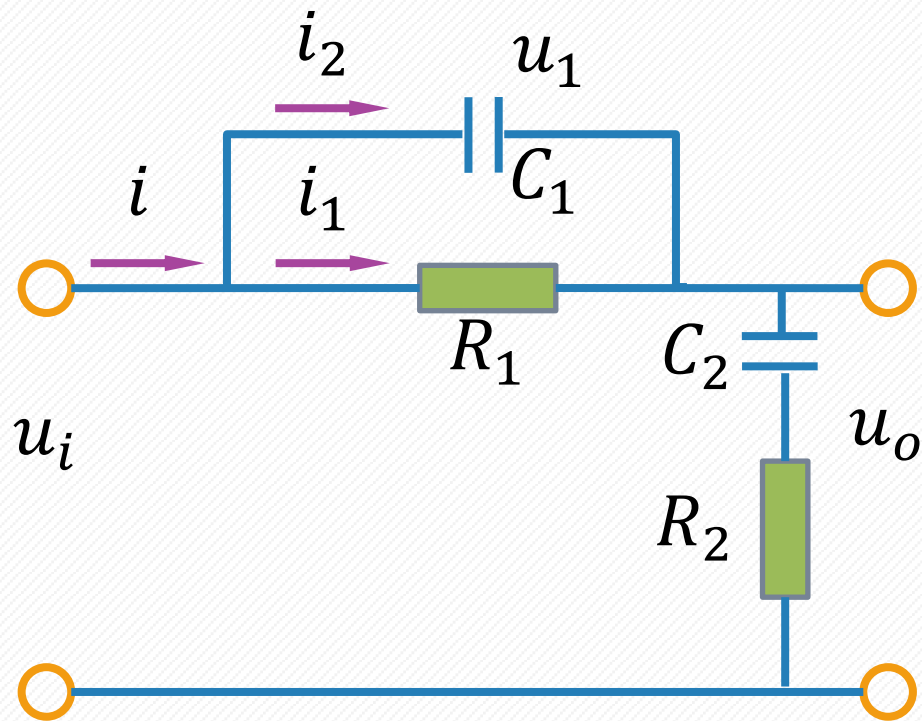
单独使用超前或滞后网络均无法达成设计目标时

滞后-超前联合校正

$$G_c(s) = \frac{U_o}{U_i} = \frac{T_a s + 1}{a T_a s + 1} \cdot \frac{T_b s + 1}{\frac{T_b}{a} s + 1}$$

$$T_a = R_1 C_1$$

$$T_b = R_2 C_2$$



滞后-超前网络特性

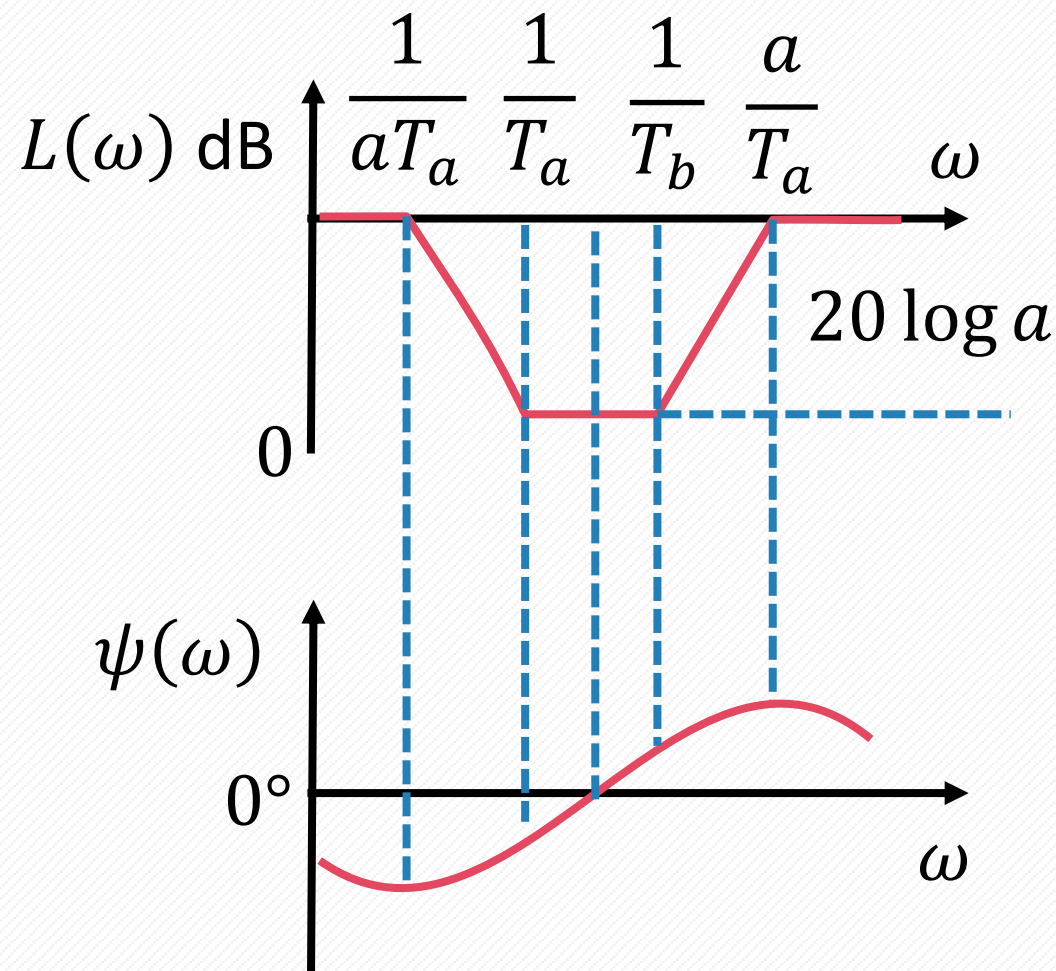
$$G_c(s) = \underbrace{\frac{T_a s + 1}{a T_a s + 1}}_{\text{滞后部分}} \cdot \underbrace{\frac{T_b s + 1}{\frac{T_b}{a} s + 1}}_{\text{超前部分}} \quad (a > 1)$$

滞后-超前网络特点

幅值衰减

相角超前

若高频与低频部分衰减因子分别为 $\alpha > 1$, $\beta < 1$,
则高频部分幅值为 $20\log \alpha\beta$



步骤

给定 e_{ss}^* , ω_c^* , γ^*

实质：综合利用滞后网络幅值衰减、超前网络相角超前的特性，改造开环频率特性，提高系统性能

1

由 $e_{ss}^* \rightarrow K$

2

由 $G_0(s) \rightarrow L_0(\omega) \rightarrow \omega_{c0} \rightarrow \gamma_0$: 超前、滞后均无效时

3

确定 $\psi_m = \gamma^* - \gamma_0(\omega_c) + 6^\circ$

$$a = \frac{1 + \sin \psi_m}{1 - \sin \psi_m}$$

4

设计超前校正 G_{lead}

$$\omega_m = \omega_c, \quad \omega_4 = \omega_m \sqrt{a}, \quad \omega_3 = \omega_4 / a$$

5

根据超前后的系统在 ω_m 处的幅值决定滞后网络系数

$$L_{lead}(\omega_m) = -20 \log b$$

6

设计滞后校正 G_{lag}

$$\omega_2 = \omega_c / 10, \quad \omega_1 = \omega_2 b$$

7

$G(s) = G_{lag}(s)G_{lead}(s)G_0(s)$ 验算 ω_c , γ 是否满足要求

例1 超前网络设计

$$G_0(s) = \frac{K}{s \left(\frac{s}{10} + 1 \right) \left(\frac{s}{60} + 1 \right)}$$

要求: 1) $K = 100$, 2) $\gamma = 50^\circ$, 3) $h^* \geq 10 \text{ dB}$, 4) $\omega_c = 20$

$$|G_0(j\omega_{c0})| = \frac{100}{\omega_{c0} \sqrt{\left(\frac{\omega_{c0}}{10}\right)^2 + 1} \sqrt{\left(\frac{\omega_{c0}}{60}\right)^2 + 1}} \approx \frac{100}{\omega_{c0} \sqrt{\left(\frac{\omega_{c0}}{10}\right)^2 + 1}} = 1$$

$$\omega_{c0} \approx 31 \text{ rad/s} \quad \gamma_0(\omega_{c0}) = 180^\circ - 90^\circ - \arctan \frac{31}{10} - \arctan \frac{31}{60} = -9.4^\circ$$

$$\psi_m = \gamma^* - \gamma_0(\omega_c) + 6^\circ = 50^\circ - 180^\circ + 90^\circ + \arctan \frac{20}{10} + \arctan \frac{20}{60} + 6^\circ = 47.9^\circ$$

$$a = \frac{1 + \sin \psi_m}{1 - \sin \psi_m} = 6.7512 \quad \omega_4 = \omega_m \sqrt{a} = \omega_c \sqrt{a} = 52 \quad \omega_3 = \omega_4 / a = 7.7$$

例1 超前网络设计

$$G_0(s) = \frac{K}{s \left(\frac{s}{10} + 1 \right) \left(\frac{s}{60} + 1 \right)}$$

要求: 1) $K = 100$, 2) $\gamma = 50^\circ$, 3) $h^* \geq 10$ dB, 4) $\omega_c = 20$

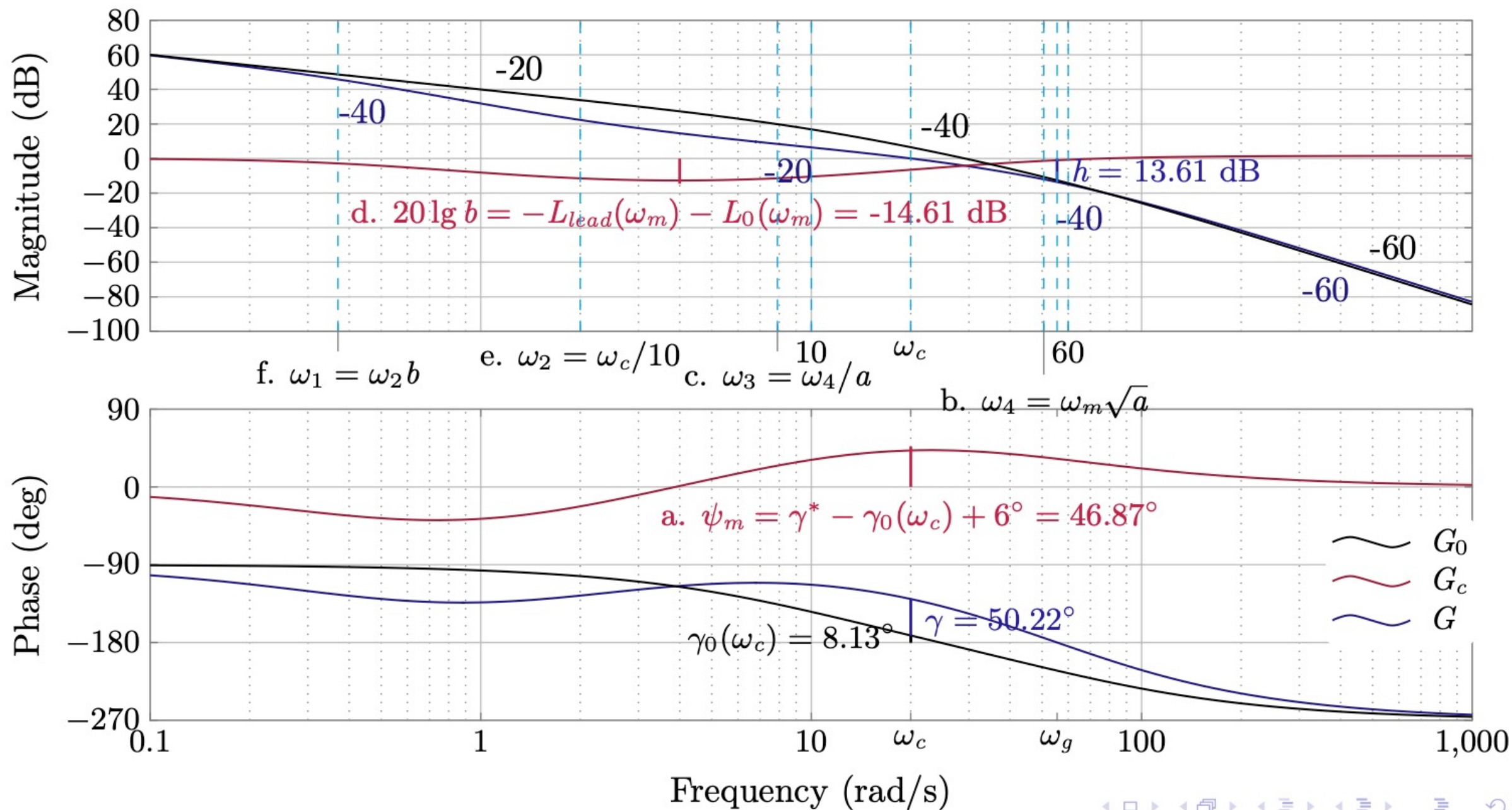
$$\omega_4 = \omega_m \sqrt{a} = \omega_c \sqrt{a} = 52 \quad \omega_3 = \omega_4 / a = 7.7$$

$$20 \log b = -L_{\text{lead}}(\omega_m) = 20 \log \frac{100 \sqrt{\left(\frac{\omega_m}{7.7} \right)^2 + 1}}{\omega_c 0 \sqrt{\left(\frac{\omega_m}{10} \right)^2 + 1} \sqrt{\left(\frac{\omega_m}{60} \right)^2 + 1} \sqrt{\left(\frac{\omega_m}{52} \right)^2 + 1}} = -14.82 \text{ dB}$$

$$b = 0.18 \quad \omega_2 = \omega_c / 10 = 2 \quad \omega_1 = \omega_2 b = 0.36$$

$$G(s) = G_{\text{lag}}(s) G_{\text{lead}}(s) G_0(s) = \frac{\frac{s}{2} + 1}{\frac{s}{0.36} + 1} \frac{\frac{s}{7.7} + 1}{\frac{s}{52} + 1} \frac{100}{s \left(\frac{s}{10} + 1 \right) \left(\frac{s}{60} + 1 \right)}$$

$$\gamma(\omega_c) = 180^\circ - \arctan \frac{20}{0.36} + \arctan \frac{20}{2} + \arctan \frac{20}{7.7} - \arctan \frac{20}{52} - 90^\circ - \arctan \frac{20}{10} - \arctan \frac{20}{60} = 51.36 \text{ 满足}$$



比例-积分-微分 (PID) 调节器

无需知道精确的控制对象模型

$$u(t) = K_P e(t) + K_I \int_0^t e(\tau) d\tau + K_D \frac{de(t)}{dt}$$

$$G_c(s) = \frac{U(s)}{E(s)} = K_P + \frac{K_I}{s} + K_D s = \frac{K_D s^2 + K_P s + K_I}{s}$$

常采用试探法确定PID参数

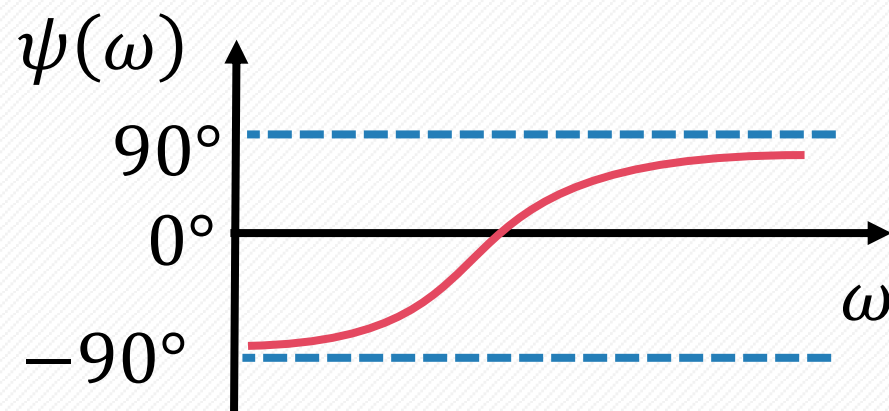
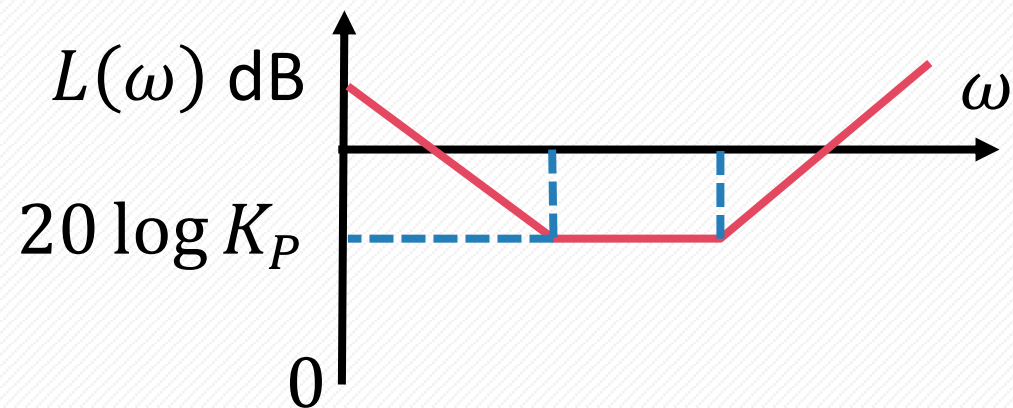
比例-积分-微分 (PID) 调节器

$$G_c(s) = \frac{K_D s^2 + K_P s + K_I}{s}$$

积分校正低频，微分校正高频

PID 调节器是一种滞后-超前校正器，同样会放大高频噪声，需要引入极点：

$$G_c(s) = K_P + \frac{K_I}{s} + \frac{K_D s}{s - p_{add}}$$



PID整定 (Tuning)

[Astrom]

类型	K_P	K_I	K_D
P	$1/a$		
PI	$0.9/a$	3τ	
PID	$1.2/a$	2τ	0.5τ

阶跃响应法

τ : 近似延迟时间

a/τ : 最陡处斜率

类型	K_P	K_I	K_D
P	$0.5K_C$		
PI	$0.4K_C$	$0.8T_C$	
PID	$0.6K_C$	$0.5T_C$	$0.125T_C$

频率响应法

K_C : 临界振荡增益

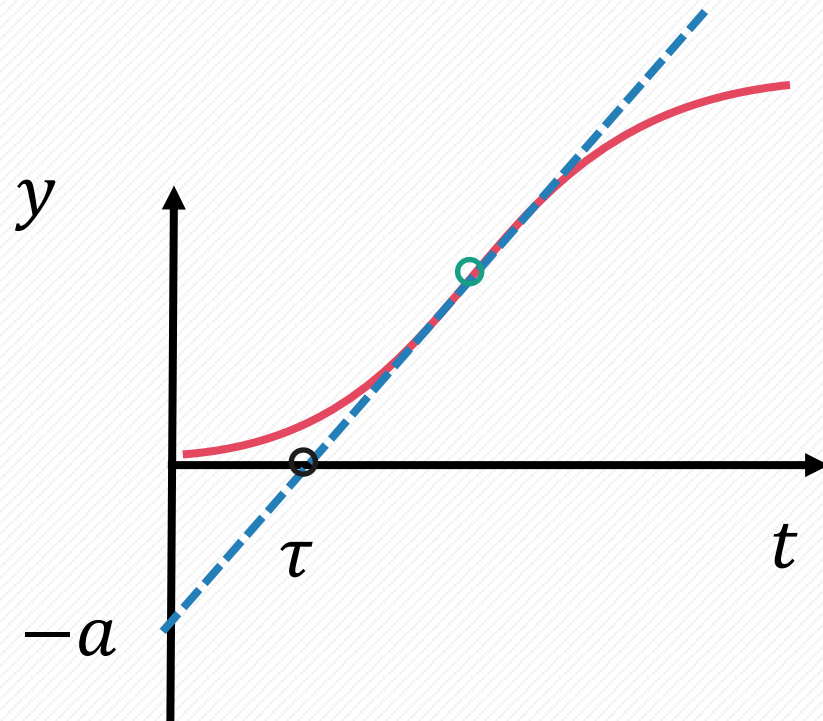
T_C : 临界振荡增益

PID整定 (Tuning)

类型	K_P	K_I	K_D
P	$1/a$		
PI	$0.9/a$	3τ	
PID	$1.2/a$	2τ	0.5τ

τ : 近似延迟时间

a/τ : 最陡处斜率



PID整定 (Tuning)

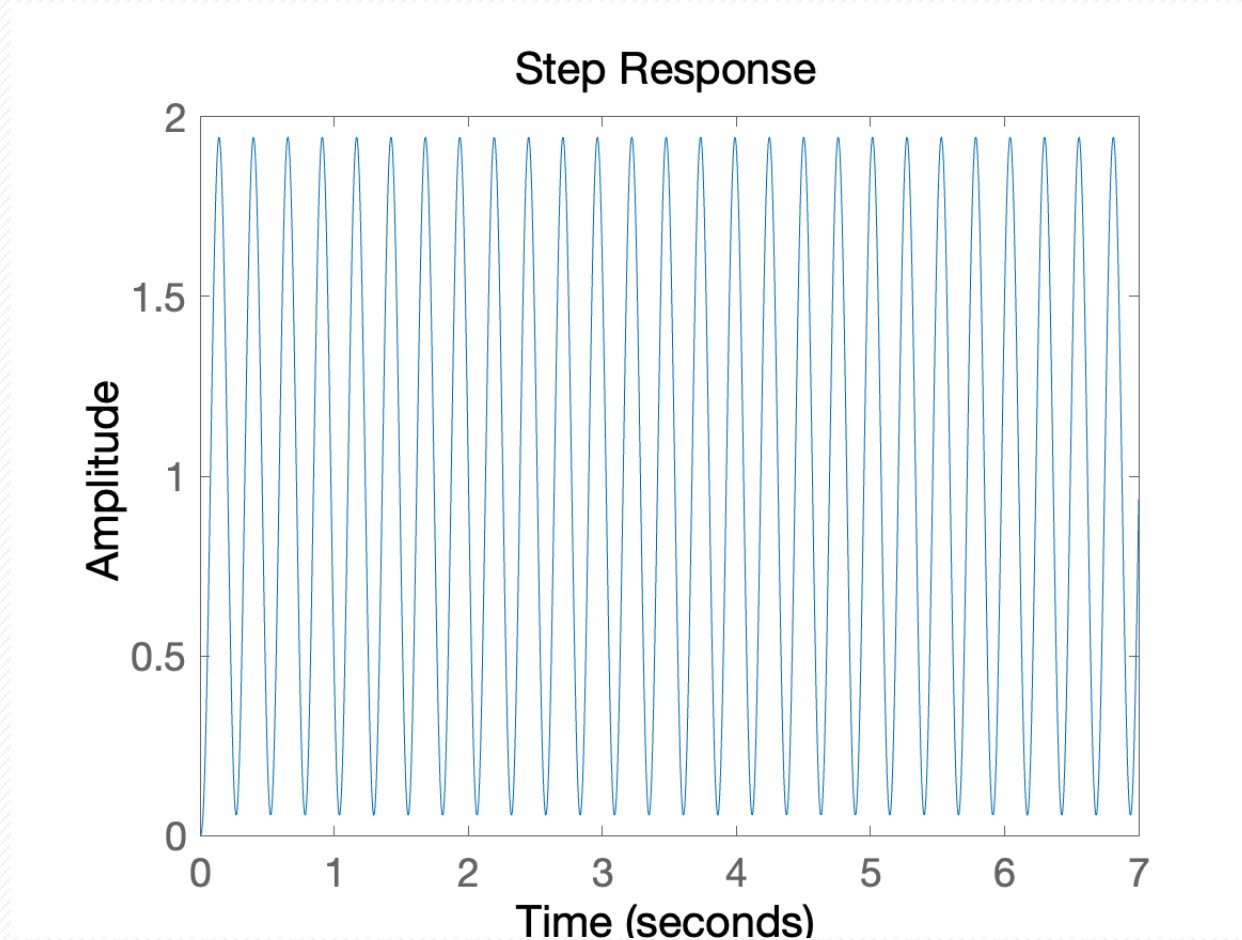
原系统闭环不稳定，无法应用阶跃响应法

类型	K_P	K_I	K_D
P	$0.5K_C$		
PI	$0.4K_C$	$0.8T_C$	
PID	$0.6K_C$	$0.5T_C$	$0.125T_C$

$$K_C = 0.7$$

$$T_C = 0.2566$$

$$G_0(s) = \frac{K}{s \left(\frac{s}{10} + 1 \right) \left(\frac{s}{60} + 1 \right)}$$



PID整定 (Tuning)

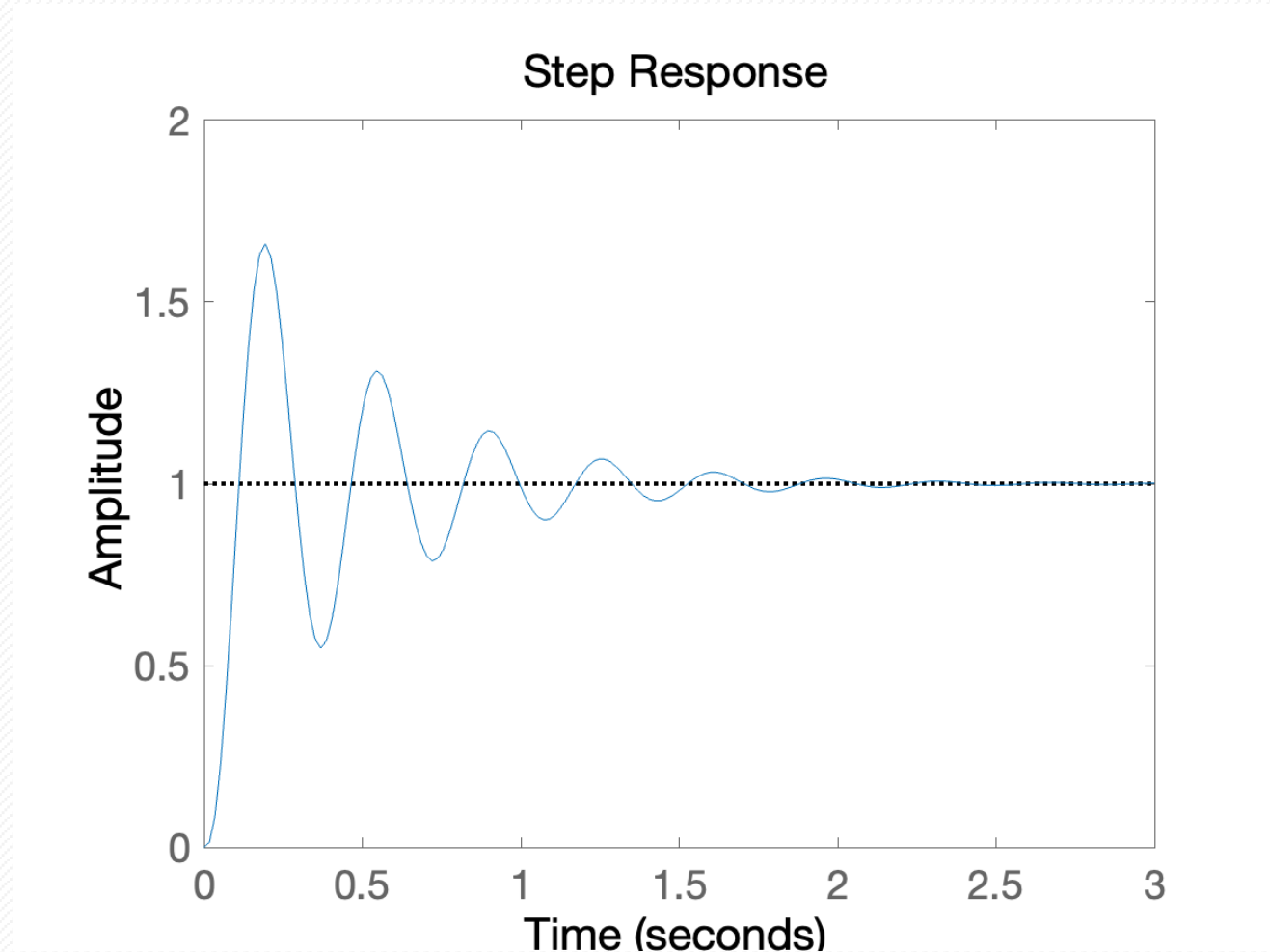
类型	K_P	K_I	K_D
P	$0.5K_C$		
PI	$0.4K_C$	$0.8T_C$	
PID	$0.6K_C$	$0.5T_C$	$0.125T_C$

$$K_C = 0.7$$

$$T_C = 0.2566$$

$$\text{P控制: } K_P = 0.5K_C = 0.35$$

$$G_0(s) = \frac{K}{s \left(\frac{s}{10} + 1 \right) \left(\frac{s}{60} + 1 \right)}$$



PID整定 (Tuning)

类型	K_P	K_I	K_D
P	$0.5K_C$		
PI	$0.4K_C$	$0.8T_C$	
PID	$0.6K_C$	$0.5T_C$	$0.125T_C$

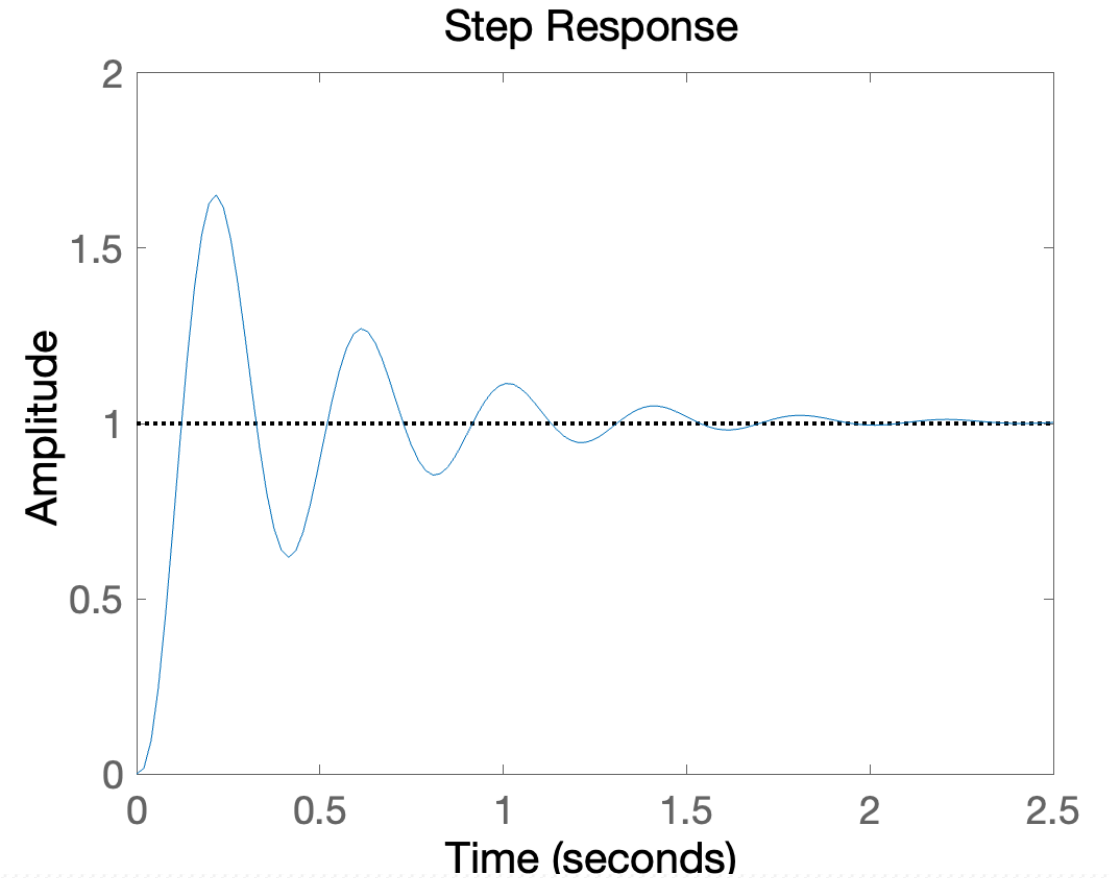
$$K_C = 0.7$$

$$T_C = 0.2566$$

$$\text{PI控制: } K_P = 0.4K_C = 0.28$$

$$K_I = 0.8T_C = 0.21$$

$$G_0(s) = \frac{K}{s \left(\frac{s}{10} + 1 \right) \left(\frac{s}{60} + 1 \right)}$$



PID整定 (Tuning)

类型	K_P	K_I	K_D
P	$0.5K_C$		
PI	$0.4K_C$	$0.8T_C$	
PID	$0.6K_C$	$0.5T_C$	$0.125T_C$

$$K_C = 0.7$$

$$T_C = 0.2566$$

$$\text{PID控制: } K_P = 0.6K_C = 0.42$$

$$K_I = 0.5T_C = 0.13$$

$$K_D = 0.125T_C = 0.03$$

$$G_0(s) = \frac{K}{s \left(\frac{s}{10} + 1 \right) \left(\frac{s}{60} + 1 \right)}$$

